Relativistic harmonic oscillator: (motion in 1d)

$$\gamma \frac{d}{dt}\left(\frac{mv}{\sqrt{1-v^2/c^2}}\right) = \gamma(-kx) = \gamma(-m\omega_0^2 x)$$

<u>Make a point:</u> in general $\frac{d}{dt}\frac{m\vec{v}}{\sqrt{1-v^2/c^2}} = \frac{m}{\sqrt{1-v^2/c^2}}\frac{d\vec{v}}{dt} +$

$$+ m\vec{v}\,\frac{\frac{v}{c^2}\frac{dv}{dt}}{\left(1-\frac{v^2}{c^2}\right)^{3/2}} = m\gamma\frac{d\vec{v}}{dt} + \underbrace{m\gamma^3\frac{v}{c^2}\frac{dv}{dt}\vec{v}}_{} = \vec{F}$$

points along $\vec{v}$
$\Rightarrow$ force is not lined up with acceleration $\frac{d\vec{v}}{dt}$ in general!

In $d=1$, $\vec{v} = v = \frac{dx}{dt}$

$$\Rightarrow \quad m\gamma\left(1+\gamma^2\frac{v^2}{c^2}\right)\frac{dv}{dt} = \vec{F} = -kx$$

To leading order in $\frac{v^2}{c^2} \ll 1$:

$$\frac{m}{\sqrt{1-\frac{v^2}{c^2}}}\left(1+\frac{v^2/c^2}{1-v^2/c^2}\right)\frac{dv}{dt} = -kx$$

$$m\left(1+\frac{v^2}{2c^2}\right)\left(1+\frac{v^2}{c^2}\right)\frac{dv}{dt} = -kx$$

$$m\left(1+\frac{3v^2}{2c^2}\right)\frac{dv}{dt} = -kx = -m\omega_0^2 x$$

$$\frac{dv}{dt} + \omega_0^2 x = -\frac{3v^2}{2c^2}\frac{dv}{dt}$$

$$v = \frac{dx}{dt} \Rightarrow \quad \ddot{x} + \omega_0^2 x = \underbrace{-\frac{3}{2c^2}\dot{x}^2\ddot{x}}_{\text{leading relativistic correction}}$$

Solve by iterations: $x = x^{(0)} + x^{(1)}$ where $x^{(1)}$ represents relativistic correction

$$\ddot{x}^{(0)} + \ddot{x}^{(1)} + \omega_0^2\left(x^{(0)} + x^{(1)}\right) = -\frac{3}{2c^2}\left(\dot{x}^{(0)}\right)^2 \ddot{x}^{(0)}$$

$$\Rightarrow \begin{cases} \left(\ddot{x}^{(0)} + \omega_0^2 x^{(0)}\right) = 0 \\ \ddot{x}^{(1)} + \omega_0^2 x^{(1)} = -\frac{3}{2c^2}\left(\dot{x}^{(0)}\right)^2 \ddot{x}^{(0)} \end{cases}$$

$$\underline{x^{(0)} = A_0 \cos\omega t}, \quad x^{(1)} = A_1 \cos 3\omega t$$

$$\dot{x}^{(0)} = -A_0\omega \sin\omega t$$

$$\ddot{x}^{(0)} = -A_0\omega^2 \cos\omega t$$

$$\Rightarrow \text{RHS} = +\frac{3}{2c^2} A_0^3 \omega^4 \underbrace{\sin^2\omega t \cos\omega t}$$

$$-\frac{1}{8}\left(e^{i\omega t} - e^{-i\omega t}\right)^2\left(e^{i\omega t} + e^{-i\omega t}\right) =$$

$$= -\frac{1}{8}\left(e^{i2\omega t} + e^{-i2\omega t} - 2\right)\left(e^{i\omega t} + e^{-i\omega t}\right)$$

$$= -\frac{1}{8}\left(e^{i3\omega t} + e^{i\omega t} + e^{-i\omega t} + e^{-i\omega t} - 2\left(e^{i\omega t} + e^{-i\omega t}\right)\right) =$$

$$= \frac{1}{4}\left(\cos\omega t - \cos 3\omega t\right)$$

$$\text{RHS} = \frac{3\omega^4}{8c^2} A_0^3 \left(\cos\omega t - \cos 3\omega t\right)$$

LHS: $\ddot{x}^{(1)} = -9\omega^2 A_1 \cos 3\omega t$

$$\Rightarrow \quad -9\omega^2 A_1 \cos 3\omega t + \omega_0^2 A_1 \cos 3\omega t$$

$$+ \left(-A_0 \omega^2 \cos \omega t + \omega_0^2 A_0 \cos \omega t\right)$$

$$\Rightarrow \quad \cos\omega t \left[-A_0\omega^2 + A_0\omega_0^2\right] + \cos 3\omega t \left[\omega_0^2 A_1 - 9\omega^2 A_1\right] =$$

$$= \frac{3\omega^4}{8c^2} A_0^3 \left(\cos\omega t - \cos 3\omega t\right)$$

$$A_0 (\omega_0^2 - \omega^2) = \frac{3\omega^4}{8c^2} A_0^3 \Rightarrow \omega^2 - \omega_0^2 = -\frac{3\omega^4}{8c^2} A_0^2$$

$$A_1 (\omega_0^2 - 9\omega^2) = -\frac{3\omega^4}{8c^2} A_0^3 \Rightarrow (9\omega^2 - \omega_0^2) A_1 = \frac{3\omega^4}{8c^2} A_0^3$$

$$\Rightarrow \quad \omega = \left(\omega_0^2 - \frac{3\omega_0^4}{8c^2} A_0^2\right)^{1/2} = \omega_0 \left(1 - \frac{3\omega_0^2}{16c^2} A_0^2\right) < \omega_0$$

$\Downarrow$

period of motion has increased.

$$\Rightarrow \quad 9\omega^2 - \omega_0^2 = 9\left(\omega_0^2 - \frac{3\omega_0^4}{8c^2} A_0^2\right) - \omega_0^2 =$$

$$= 8\omega_0^2 - \frac{27\omega_0^4 A_0^2}{8c^2}$$

$$\Rightarrow \left(8\omega_0^2 - \frac{27\omega_0^2 A_0^2}{8c^2}\right) A_1 = \frac{3\omega_0^4}{8c^2} A_0^3$$

$$A_1 = \frac{3\omega_0^2 A_0^3}{8c^2} \frac{1}{8\left(1 - \frac{27\omega_0^2 A_0^2}{64c^2}\right)}$$

$$A_1 = \frac{3\omega_0^2 A_0^3}{64c^2} \Rightarrow \frac{A_1}{A_0} \approx \left(\frac{\omega_0 A_0}{c}\right)^2 \sim \left(\frac{V_{max}}{c}\right)^2.$$

$$x(t) = A_0 \cos\omega t + \frac{3\omega_0^2 A_0^3}{64c^2} \cos 3\omega t$$

Via Energy conservation

$$\frac{mc^2}{\sqrt{1-\frac{v^2}{c^2}}} + \underbrace{\frac{1}{2} m\omega_0^2 x^2}_{U(x)} = E = mc^2 + \frac{1}{2} m\omega_0^2 A_0^2$$

$$v^2 = \dot{x}^2 = c^2 \left[ 1 - \left( \frac{mc^2}{E - U(x)} \right)^2 \right]$$

$$\frac{dx}{dt} = c \sqrt{\frac{(E - U(x) - mc^2)(E - U(x) + mc^2)}{(E - U(x))^2}}$$

$$T = \int dt = \frac{1}{c} \underbrace{\int dx}_{4\int_0^{A_0} dx} \frac{E - U(x)}{[E - mc^2 - U(x)]^{1/2} [E + mc^2 - U(x)]^{1/2}}$$

$$= \frac{4}{c} \int_0^{A_0} dx \frac{mc^2 + \frac{1}{2} m\omega_0^2 (A_0^2 - x^2)}{\sqrt{\frac{1}{2} m\omega_0^2 (A_0^2 - x^2)} \sqrt{2mc^2 + \frac{1}{2} m\omega_0^2 (A_0^2 - x^2)}}$$

$$x = A_0 y$$

$$T = \frac{4A_0}{c} \int_0^1 dx \frac{mc^2 + \frac{m\omega_0^2 A_0^2}{2} (1 - y^2)}{\sqrt{\frac{m\omega_0^2 A_0^2}{2} (1 - y^2)} \sqrt{2mc^2 + \frac{m\omega_0^2 A_0^2}{2} (1 - y^2)}}$$

$$y = \cos\theta$$

$$T = 4A_0 c \int_0^1 dy \; \frac{1 + \frac{\omega_0^2 A_0^2}{2c^2}(1-y^2)}{\sqrt{\frac{\omega_0^2 A_0^2}{2}(1-y^2)}\;\sqrt{2}\,c\sqrt{1+\frac{\omega_0^2 A_0^2}{4c^2}(1-y^2)}}$$

$$= \frac{4A_0}{\omega_0 A_0} \int_0^1 \frac{dy}{\sqrt{1-y^2}} \; \frac{1+2\xi(1-y^2)}{\sqrt{1+\xi(1-y^2)}}$$

$$y = \cos\theta \qquad \int_0^1 \frac{dy}{\sqrt{1-y^2}} = \int_{\pi/2}^{0} \frac{-d\theta \sin\theta}{\sin\theta} = \int_0^{\pi/2} d\theta$$

$$T = \frac{4}{\omega_0} \int_0^{\pi/2} d\theta \; \frac{1+2\xi \sin^2\theta}{\sqrt{1+\xi \sin^2\theta}}$$

$$\xi \equiv \left(\frac{\omega_0 A_0}{2c}\right)^2$$

$\Rightarrow$ $\xi \ll 1$: $\quad T = \frac{4}{\omega_0} \cdot \frac{\pi}{2} = \frac{2\pi}{\omega_0}$

$\xi \gg 1$: $\quad T = \frac{4}{\omega_0} \frac{2\sqrt{\xi}}{} \underbrace{\int_0^{\pi/2} d\theta \sin\theta}_{1} =$

$$= \frac{8}{\omega_0} \frac{\omega_0 A_0}{2c} = \frac{4A_0}{c}$$

# Least action principle.

As system moves from state a to state b, it does so along the trajectory that makes the action integral

$$S = \int_a^b dt\, L$$

an extremum (generally minimum), subject to constraint that the end points a, b are fixed (e.g. $x(t_a) = x_a$, $y(t_b) = y_b$).

$$L = L(\{q_i\}, \{\dot{q}_i\}, t)$$

$q_i$ = generalized coordinates

$\dot{q}_i$ = generalized velocities

Consider that optimal $q(t)$ is known and study how action $S$ changes when $q(t) \to q(t) + \delta q(t)$ is varied. But $q(t_1) = q_1$, $q(t_2) = q_2$ are held fixed:

$$S = \int_{t_1}^{t_2} dt\, L(q, \dot{q})$$

$$\delta S = \int_{t_1}^{t_2} dt \left( L(q + \delta q, \dot{q} + \delta \dot{q}) - L(q, \dot{q}) \right)$$

$$= \int dt \left( \frac{\partial L}{\partial q} \delta q + \frac{\partial L}{\partial \dot{q}} \delta \dot{q} \right)$$

Require $\delta S = 0$ for the trajectory $q(t)$ to be optimal!

$$\frac{\partial L}{\partial \dot{q}} \delta \dot{q} = \frac{d}{dt}\left( \frac{\partial L}{\partial \dot{q}} \delta q \right) - \delta q \frac{d}{dt}\left( \frac{\partial L}{\partial \dot{q}} \right)$$

$$\Rightarrow \delta S = \int_{t_1}^{t_2} dt \left\{ \frac{d}{dt}\left( \frac{\partial L}{\partial \dot{q}} \delta q \right) + \left( \frac{\partial L}{\partial q} - \frac{d}{dt} \frac{\partial L}{\partial \dot{q}} \right) \delta q \right\}$$

$$\int_{t_1}^{t_2} dt \, \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}} \delta q\right) = \frac{\partial L}{\partial \dot{q}} \delta q \Big|_{t_1}^{t_2} = 0 \quad \text{because } \delta q(t_1) = 0 = \delta q(t_2).$$

$$\Rightarrow \delta S = \int_{t_1}^{t_2} dt \left(\frac{\partial L}{\partial q} - \frac{d}{dt}\frac{\partial L}{\partial \dot{q}}\right) \delta q$$

Since $\delta q(t)$ is arbitrary, we must require that $\delta S = 0$ leads to Euler-Lagrange equation:

$$\frac{\partial L}{\partial q} - \frac{d}{dt}\frac{\partial L}{\partial \dot{q}} = 0.$$

$$p_j = \frac{\partial L}{\partial \dot{q}_j} = \text{generalized momentum.}$$

Non-relativistic mechanics: $L = \frac{1}{2} m \dot{q}^2 - U(q)$

$$\Rightarrow \frac{\partial L}{\partial q} = -\frac{\partial U}{\partial q} \Rightarrow -\vec{\nabla} U$$

$$\frac{\partial L}{\partial \dot{q}} = p = m\dot{q}$$

$$\Rightarrow \text{E-L equation:} \quad \underbrace{\frac{d}{dt}(m\dot{q})}_{m\ddot{q}} = \underbrace{-\frac{\partial U}{\partial q}}_{F}$$

Symmetries:

1) homogeneity of space : uniform translation by $\Delta\vec{R}$ should not change motion of a system of particles.

$$\Delta L = \sum_i \frac{\partial L}{\partial \vec{r}_i} \cdot \Delta\vec{R} = \Delta\vec{R} \cdot \sum_i \frac{d}{dt} \frac{\partial L}{\partial \dot{\vec{r}}_i} = 0$$

$$\Rightarrow \frac{d}{dt} \sum_i \frac{\partial L}{\partial \dot{\vec{r}}_i} = \frac{d}{dt} \sum_i \vec{p}_i = 0$$

$\Rightarrow$ total momentum is conserved!

2) uniformity of time : motion of a closed system is not affected by a shift $t \to t + \Delta t$.

This means that Lagrangian $L(q, \dot{q})$ depends on $t$ via $q(t)$ and $\dot{q}(t)$, but does not have <u>explicit</u> dependence of $t$!

Then $$\frac{dL}{dt} = \sum_i \left( \frac{\partial L}{\partial q_i} \frac{dq_i}{dt} + \frac{\partial L}{\partial \dot{q}_i} \frac{d\dot{q}_i}{dt} \right) + \underbrace{\cancel{\frac{\partial L}{\partial t}}}_{=0} =$$

$$\left( \text{use } \frac{\partial L}{\partial q_i} = \frac{d}{dt} \frac{\partial L}{\partial \dot{q}_i} \right)$$

$$= \sum_i \frac{d}{dt}\left( \frac{\partial L}{\partial \dot{q}_i} \right) \dot{q}_i + \frac{\partial L}{\partial \dot{q}_i} \ddot{q}_i =$$

$$= \frac{d}{dt} \sum_i \left( \dot{q}_i \frac{\partial L}{\partial \dot{q}_i} \right)$$

Therefore $\frac{d}{dt}\left\{\sum_i \left(\dot{q}_i \frac{\partial L}{\partial \dot{q}_i}\right) - L\right\} = 0$

$\Rightarrow \quad \mathcal{E} = \sum_i \dot{q}_i \frac{\partial L}{\partial \dot{q}_i} - L$ is integral of motion (conserved).

But for $L = \frac{1}{2} m \dot{q}^2 - U(q)$ , $\frac{\partial L}{\partial \dot{q}_i} = m\dot{q} = p$

$\Rightarrow \quad \mathcal{E} = \dot{q}(m\dot{q}) - \left(\frac{1}{2} m\dot{q}^2 - U(q)\right) = \frac{1}{2} m\dot{q}^2 + U(q)$

is just the <u>energy</u>.